



DEVELOPER AND DESIGN SPECIFICATION

Life in the UK Test Prep App

An offline-first Android study app that runs on the free Firebase Spark plan: the question bank lives in a local database on the phone, and Google Play delivers the content.

Version	1.1 – adds brand and logo, light and dark themes, six supporting screens
Date	1 October 2026
Product and content	Amila
Android and backend	Yasantha
Platform	Android (Kotlin, Jetpack Compose), Firebase Spark, Google Play
Status	Ready for design and development. Open decisions are listed in section 25.

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What changed in v1.1 (1 October 2026). Parts A, B and D are unchanged.

- **Logo:** clock tower rising from an open book, with app icon rules (18, 19.4).
- **Colour:** brand colours and one colour per chapter added (19.2, 19.3); background, surfaceVariant and outline warmed slightly; success and error containers named (19.1).
- **Themes:** Light, Dark or System picker in Settings, System by default (19.7, S20).
- **Designs:** section 21 now shows high-fidelity screens instead of wireframes; flow updated (20).
- **Screens:** six supporting screens added – S05b, S11b, S13b, S14b, S15b, S16b (22).
- **Handoff:** components and checklist cover all 26 screens in both themes (23).

Part A – Product

What we are building, for whom, and how it makes money.

1 Summary and key decisions

We are building a paid Android app that helps adults pass the Life in the UK Test first time. All study runs offline on the phone, content ships inside the app, and the backend costs nothing to run until the app has thousands of daily users.

£0
monthly hosting on the Firebase Spark plan; no payment card linked

~70–100 MB
on the phone including audio; ~20 MB without the audio pack

6 weeks
to a production release on Google Play, evenings and weekends

Decision	Choice
Platform	Android first (Kotlin, Jetpack Compose, Material 3). iOS is out of scope for v1.
Content delivery	Question bank bundled as a read-only SQLite file (<code>content.db</code>); audio delivered by Play Asset Delivery. No Cloud Storage.
User data	Kept on the phone in <code>user.db</code> . Optional sync to Firestore in London (<code>eu-west2</code>), one write per study session.
Firebase plan	Spark (free, no card). Auth, Firestore, Remote Config, Crashlytics, Analytics.
Content updates	Shipped as normal app updates. Remote Config can hide a faulty question immediately.
Monetisation	Hard paywall after a free sample, 7-day free trial, monthly and annual plans through Google Play Billing and RevenueCat.
AI usage	At build time only, to draft questions and explanations that Amila reviews. No AI calls from the app in v1.
Publisher	Yasantha's Google Play developer account (to be confirmed in the partnership agreement).

2 The exam

The Life in the UK Test is a computer-based multiple-choice test that adults take for settlement or British citizenship. Every design decision in the app maps to these facts.

Fact	Value	What it means for the app
Format	24 questions in 45 minutes	Mock exams copy this exactly, including the countdown timer.
Pass mark	75% (18 of 24)	Results and the readiness score use 18/24 as the pass line.

Fact	Value	What it means for the app
Fee	£50 per attempt; unlimited retakes, each paid	Strong reason to pay a few pounds to pass first time.
Source material	Life in the United Kingdom: A Guide for New Residents, 3rd edition	Every question carries a handbook chapter and section reference.
Exempt	Under 18, 65 and over, some long-term conditions	Audience is adults, often non-native English speakers.
Future changes	The 2025 Immigration White Paper proposed a new test; no new handbook or date has been published	<code>content.db</code> is versioned and replaceable so a new syllabus ships as a normal update.

Market reality. Free websites (for example one with 40 practice tests and 17 mock exams) and an official practice app already exist. A paid app wins on experience: offline study, spaced repetition, a readiness score, audio, and plain-English explanations for non-native speakers.

3 MVP scope

In scope for v1.0

#	Feature	Description
1	Onboarding	Test date, daily goal, reminder time, then a 10-question free sample before the paywall.
2	Topic practice	Questions by handbook chapter and section, instant feedback and an explanation after each answer.
3	Mixed practice	20 random questions across all chapters.
4	Mock exams	24 questions, 45 minutes, no feedback until the end, results by chapter, full answer review.
5	Spaced repetition	Wrong and flagged questions come back on a schedule (section 9.1).
6	Readiness score	0–100% from recent mocks and coverage (section 9.2).
7	Study notes	Original plain-English summaries per section, with key-fact boxes.
8	Audio	Read-aloud for every question and note.
9	Study timer	Focus sessions (reusing Daily Focus code) that end with a 5-question quiz.
10	Streaks and reminders	One daily notification at the user's chosen time.
11	Subscription	Hard paywall, 7-day trial, restore purchases, manage in Google Play.
12	Optional backup	Link a Google account to back up and restore progress across devices.
13	Report a question	Users can flag a question as wrong or unclear.

Out of scope for v1.0

- Live AI features such as "explain differently" (candidate for v1.1, would need the Blaze plan or a separate proxy).
- Downloadable content packs from Cloud Storage.
- iOS and web versions, leaderboards, social features, tutor dashboards.
- Languages other than English in the interface (translations are a v1.2 candidate).

4 Monetisation

A hard paywall with a 7-day free trial, priced so the annual plan is the obvious choice. Prices below are starting points to check against competitors before launch.

Plan	Product ID	Price (GBP)	Notes
Monthly	litu_pro_monthly	£7.99	7-day free trial; anchor price
Annual	litu_pro_annual	£34.99	7-day free trial; highlighted as best value
Exam pass (optional test)	litu_pass_3m	£14.99	One-off, 3 months access, non-renewing

- **Free tier:** the 10-question sample and Mock 1. No ads anywhere.
- **Paywall placement:** end of onboarding (after the sample), and whenever a free user taps locked content.
- **Entitlement name:** `pro`, granted by any of the products above.
- **Google's fee:** 15% of subscription revenue.

Part B – Engineering

Architecture, data, algorithms, pipelines and release process for Yasantha and anyone joining later.

5 Architecture overview

The phone does all the work. Firebase on the free Spark plan only handles sign-in, optional backup, feature flags and crash reports. Google Play distributes the app, its content and its audio, and handles payments.

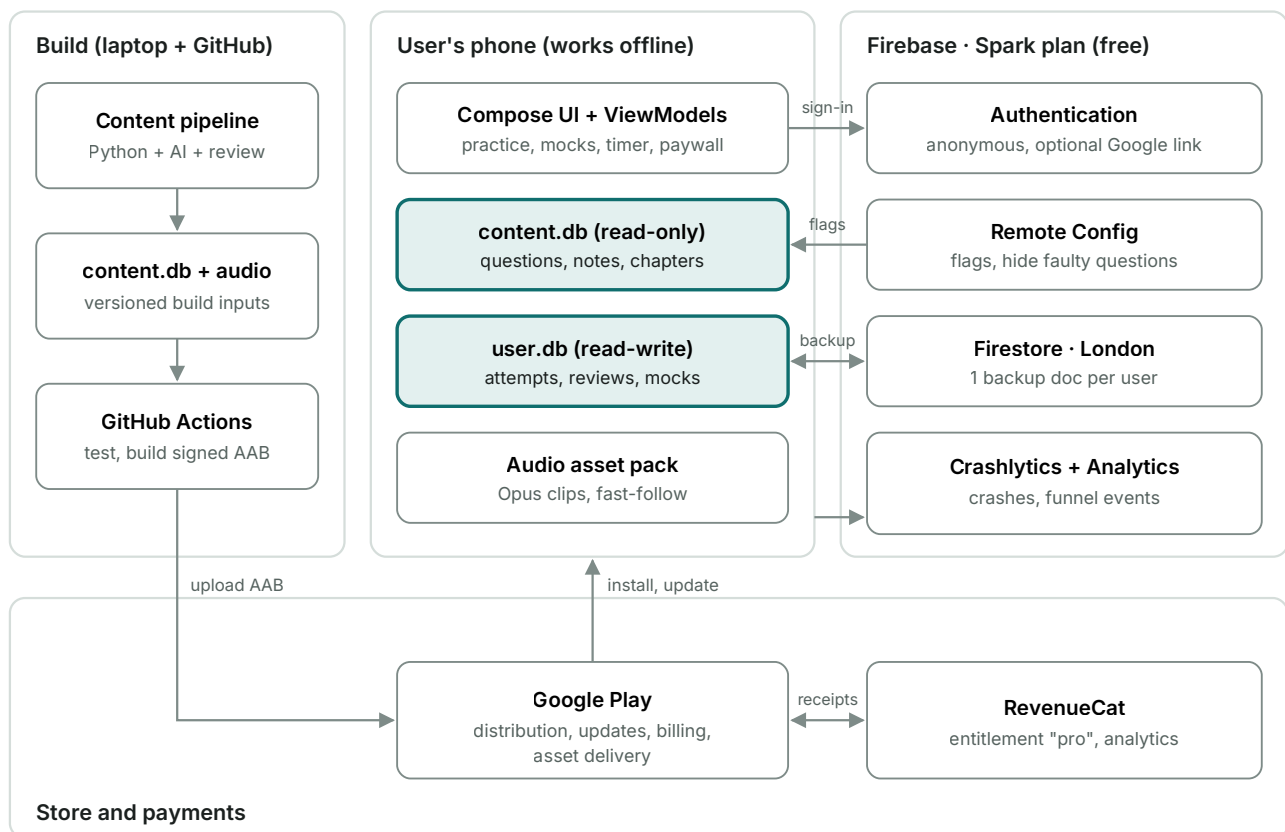


Figure 1 – Deployment architecture on the free Spark plan

Component	Runs where	Responsibility
Android app	User's phone	All study logic, scoring, scheduling, timer, audio playback, paywall.
content.db	Phone (bundled in app)	Chapters, sections, questions, options, explanations, notes. Replaced on each content release.
user.db	Phone	Attempts, review schedule, mocks, daily stats. Never replaced by an update.
Audio asset pack	Phone (delivered by Play)	One Opus clip per question and per note section.

Component	Runs where	Responsibility
Firebase Auth	Google (global)	Anonymous user on first launch; optional Google account link for backup.
Cloud Firestore	Google, europe-west2 London	One backup document per user, plus question reports.
Remote Config	Google (global)	Feature flags, minimum app version, hidden question IDs, banner text.
Crashlytics, Analytics	Google (global)	Crash reports and funnel events (section 12.6).
Google Play	Google	Distribution, updates, Play Asset Delivery, Play Billing.
RevenueCat	RevenueCat cloud	Receipt validation, entitlement state, revenue dashboard.

6 Technology stack

Use the latest stable version of each library at project start and pin versions in a Gradle version catalog (`libs.versions.toml`).

Layer	Choice	Notes
Language	Kotlin, coroutines, Flow	No Java in new code.
UI	Jetpack Compose, Material 3, Navigation Compose	Single activity.
Architecture	MVVM with unidirectional data flow	ViewModel exposes one <code>StateFlow<UiState></code> per screen.
Dependency injection	Hilt	Modules per layer.
Local database	Room (two databases)	<code>createFromAsset</code> for content.db.
Settings	Preferences DataStore	Test date, goal, reminder time, theme, audio options.
Background work	WorkManager	Daily reminder, deferred backup sync.
Audio	Media3 ExoPlayer; Android TextToSpeech fallback	Opus files from the asset pack.
Asset delivery	Play Asset Delivery (fast-follow pack)	Pack name <code>audio_pack</code> .
Firebase	Firebase BoM: Auth, Firestore, Remote Config, Crashlytics, Analytics, App Check (Play Integrity)	Spark plan only.
Billing	RevenueCat Purchases SDK (uses Play Billing Library)	Free below RevenueCat's revenue threshold.
Quality	ktlint or detekt, JUnit, Turbine, Compose UI tests, Roborazzi screenshot tests	Run in CI.
Performance	R8, Baseline Profiles, Macrobenchmark	Required for smooth budget phones.

Layer	Choice	Notes
Content tooling	Python 3, an AI API, Google Sheets for review, FFmpeg	Lives in the same repository under <code>content/</code> .

SDK levels: `minSdk 26` (Android 8.0), `targetSdk` and `compileSdk` set to the level Google Play requires at release time.

7 Project and module structure

A modular Gradle project keeps features independent and builds fast. Feature modules depend only on `core` modules, never on each other.

```

litu-app/
├── app/                Application, MainActivity, NavHost, Hilt setup
├── core/
│   ├── model/         Pure Kotlin data classes (Question, Attempt, Mock...)
│   ├── content/       content.db: Room DB, DAOs, ContentRepository
│   ├── progress/      user.db: Room DB, DAOs, ProgressRepository
│   ├── domain/        Use cases: scheduler, readiness, mock builder
│   ├── sync/          Firestore backup, Auth, WorkManager sync job
│   ├── billing/       RevenueCat wrapper, EntitlementRepository
│   ├── config/        Remote Config wrapper, feature flags
│   ├── audio/         ExoPlayer wrapper, asset pack manager, TTS fallback
│   ├── designsystem/  Theme, tokens, shared components
│   └── analytics/     Event logger interface + Firebase implementation
├── feature/
│   ├── onboarding/    home/   practice/  review/  mock/
│   ├── notes/         progress/ timer/   paywall/ settings/
│   ├── audio_pack/    Play Asset Delivery pack (fast-follow)
│   ├── content/       Python pipeline, prompts, review exports
│   │   ├── pipeline/  generate.py, validate.py, build_db.py, tts.py
│   │   └── releases/  content.db per version + changelog
│   └── .github/workflows/ ci.yml, release.yml

```

Layering rules

- **UI layer** (feature modules): Composables are stateless; ViewModels call use cases and expose `UiState` .
- **Domain layer** (`core/domain`): pure Kotlin, no Android imports, fully unit-tested.
- **Data layer**: repositories hide Room, Firestore, RevenueCat and Remote Config behind interfaces.
- All database and network work runs on `Dispatchers.IO` ; nothing blocks the main thread.

8 Local data design

Two separate SQLite databases. `content.db` is read-only and replaced with every content release; `user.db` holds the user's progress and is never wiped by an update.

8.1 content.db (read-only, bundled)

```

CREATE TABLE meta (
  key   TEXT PRIMARY KEY,           -- content_version, schema_version,
  value TEXT NOT NULL               -- built_at, question_count, handbook_edition
);

CREATE TABLE chapter (
  id       TEXT PRIMARY KEY,         -- 'CH3'
  number   INTEGER NOT NULL,         -- 1..5
  title    TEXT NOT NULL,
  in_test  INTEGER NOT NULL DEFAULT 1 -- 0 = study only (e.g. chapter 1)
);

CREATE TABLE section (
  id          TEXT PRIMARY KEY,       -- 'CH3-TUD'
  chapter_id  TEXT NOT NULL REFERENCES chapter(id),
  title       TEXT NOT NULL,
  sort        INTEGER NOT NULL
);

CREATE TABLE question (
  id          TEXT PRIMARY KEY,       -- 'Q-CH3-TUD-014' (stable forever)
  section_id  TEXT NOT NULL REFERENCES section(id),
  type        TEXT NOT NULL CHECK (type IN ('single','multi','truefalse')),
  stem        TEXT NOT NULL,
  explanation TEXT NOT NULL,         -- 1-3 sentences, plain English (≈B1)
  handbook_ref TEXT NOT NULL,        -- 'Chapter 3 · The Tudors'
  difficulty  INTEGER NOT NULL CHECK (difficulty BETWEEN 1 AND 3),
  audio_key   TEXT,                 -- 'q/Q-CH3-TUD-014.opus'
  active      INTEGER NOT NULL DEFAULT 1,
  added_in    INTEGER NOT NULL      -- content_version that introduced it
);
CREATE INDEX idx_question_section ON question(section_id, active);

CREATE TABLE answer_option (
  id          TEXT PRIMARY KEY,       -- 'Q-CH3-TUD-014-A'
  question_id TEXT NOT NULL REFERENCES question(id),
  label       TEXT NOT NULL,
  is_correct  INTEGER NOT NULL,
  sort        INTEGER NOT NULL
);
CREATE INDEX idx_option_question ON answer_option(question_id);

CREATE TABLE note (
  section_id TEXT PRIMARY KEY REFERENCES section(id),
  body_md    TEXT NOT NULL,          -- original summary, Markdown subset
  key_facts  TEXT NOT NULL,         -- JSON array of short facts
  audio_key  TEXT
);

```

Question types: `single` has 4 options and 1 correct; `multi` has 4 options and exactly 2 correct ("choose 2"); `truefalse` has 2 options. Option order is shuffled at display time except for true/false.

8.2 user.db (read-write)

```

CREATE TABLE attempt (
  id          INTEGER PRIMARY KEY AUTOINCREMENT,
  question_id TEXT    NOT NULL,          -- refers to content.db by ID only
  selected    TEXT    NOT NULL,          -- JSON array of option IDs
  correct     INTEGER NOT NULL,
  mode        TEXT    NOT NULL,          -- 'sample','practice','review','mock','timer'
  session_id  TEXT    NOT NULL,
  answered_at INTEGER NOT NULL,          -- epoch ms
  time_ms     INTEGER NOT NULL
);
CREATE INDEX idx_attempt_question ON attempt(question_id, answered_at);

CREATE TABLE review_state (
  question_id TEXT PRIMARY KEY,
  ease         REAL    NOT NULL DEFAULT 2.5,
  interval_days INTEGER NOT NULL DEFAULT 0,
  due_at       INTEGER NOT NULL,          -- epoch ms
  reps         INTEGER NOT NULL DEFAULT 0,
  lapses       INTEGER NOT NULL DEFAULT 0,
  flagged      INTEGER NOT NULL DEFAULT 0,
  updated_at   INTEGER NOT NULL
);
CREATE INDEX idx_review_due ON review_state(due_at);

CREATE TABLE study_session (
  id TEXT PRIMARY KEY, mode TEXT NOT NULL,
  started_at INTEGER NOT NULL, ended_at INTEGER,
  answered INTEGER NOT NULL DEFAULT 0, correct INTEGER NOT NULL DEFAULT 0,
  synced INTEGER NOT NULL DEFAULT 0
);

CREATE TABLE mock_exam (
  id TEXT PRIMARY KEY,
  started_at INTEGER NOT NULL, finished_at INTEGER,
  question_ids TEXT NOT NULL,          -- JSON array of 24 IDs, in order
  answers TEXT NOT NULL,              -- JSON map questionId -> optionIds
  score INTEGER, passed INTEGER,
  chapter_breakdown TEXT,             -- JSON map chapterId -> [correct,total]
  content_version INTEGER NOT NULL
);

CREATE TABLE daily_stat (
  day TEXT PRIMARY KEY,                -- 'YYYY-MM-DD' local date
  answered INTEGER NOT NULL DEFAULT 0, correct INTEGER NOT NULL DEFAULT 0,
  study_minutes INTEGER NOT NULL DEFAULT 0, goal_met INTEGER NOT NULL DEFAULT 0
);

```

8.3 Versioning and migration rules

Rule	Detail
Stable IDs	A question ID never changes and is never reused. Typo fixes keep the ID; user history stays attached.
Retiring questions	Set <code>active = 0</code> in the next release; never delete rows. Inactive questions are excluded from practice, mocks and reviews.

Rule	Detail
Content upgrade	On launch, compare bundled <code>meta.content_version</code> with the installed one. If higher, Room rebuilds content.db from the asset (<code>fallbackToDestructiveMigration</code> on the content database only).
Schema changes in user.db	Always written as explicit Room migrations with tests. Destructive migration is forbidden on user.db.
Orphan handling	user.db rows for unknown question IDs are kept but ignored, so a rollback cannot lose data.
Hidden questions	Remote Config key <code>hidden_question_ids</code> hides faulty questions until the next content release.

8.4 Settings (DataStore)

`test_date` (nullable), `daily_goal` (10, 20 or 40), `reminder_time`, `reminders_enabled`, `theme` (system, light, dark), `audio_autoplay`, `audio_speed` (0.75–1.5), `onboarding_done`, `last_content_version`.

9 Core logic and algorithms

All of this lives in `core/domain` as pure Kotlin with full unit-test coverage.

9.1 Spaced repetition (simplified SM-2)

```
fun schedule(state: ReviewState, correct: Boolean, now: Instant): ReviewState =
    if (!correct) state.copy(
        ease = max(1.3, state.ease - 0.2),
        intervalDays = 1, lapses = state.lapses + 1, reps = 0,
        dueAt = now + 1.days)
    else {
        val next = when (state.reps) {
            0 -> 1; 1 -> 3; 2 -> 7
            else -> ceil(state.intervalDays * state.ease).toInt()
        }.coerceAtMost(daysUntilTest(now) ?: 60) // never schedule past the test
        state.copy(ease = min(2.8, state.ease + 0.05), reps = state.reps + 1,
            intervalDays = next, dueAt = now + next.days)
    }
```

- A `review_state` row is created the first time a question is answered wrong or flagged.
- **Mastered** = answered correctly at least 3 times in a row and `interval_days` ≥ 7 .
- The review queue shows due items, oldest first, capped at 30 per session.

9.2 Readiness score

A single 0–100% number shown on Home. It needs at least one completed mock; until then Home asks the user to take one.

```

mockPart      = weighted average of the last 3 mock scores (%), weights 0.5, 0.3, 0.2
                (use available mocks only, re-normalising weights)
coveragePart  =  $\sum$  over test chapters ( chapterShare × chapterMastery )
                chapterMastery = questions answered correctly in the last 30 days
                                ÷ active questions in that chapter
readiness     = round( 0.65 × mockPart + 0.35 × coveragePart )

```

Readiness	Label on Home	Colour
0–59%	"Keep practising"	error
60–79%	"Getting close"	warning
80–100%	"Likely to pass"	success

9.3 Mock exam builder

1. Candidate pool: active questions in chapters with `in_test = 1`, minus hidden IDs.
2. Allocate 24 slots across chapters in proportion to each chapter's share of the pool (largest-remainder rounding). The weights can be overridden with Remote Config key `mock_chapter_weights`.
3. Within each chapter, prefer questions not used in the user's last 3 mocks, then pick randomly.
4. Shuffle the final 24. Store IDs and order in `mock_exam.question_ids` so results and review are reproducible.
5. Timer: 45:00 countdown based on wall-clock time (`started_at`), so it keeps running if the app is closed. At 00:00 the exam auto-submits.

9.4 Daily goal and streak

- A day counts when `answered ≥ daily_goal`. The streak is consecutive counted days ending today or yesterday.
- Days use the phone's local date. One "freeze" per week keeps the streak through a missed day (shown as a snowflake chip).

10 Content pipeline

AI drafts the question bank once, Amila approves every question, and a script turns the approved set into `content.db`. Target for v1.0: about 1,000 drafts to reach 600–800 approved questions.

Nothing reaches users without Amila's approval

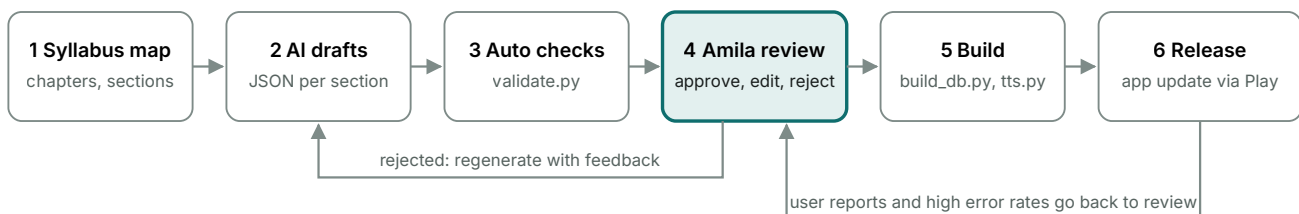


Figure 2 – Content pipeline

Draft question format (JSON, one file per section)

```
{
  "id": "Q-CH3-TUD-014",
  "section_id": "CH3-TUD",
  "type": "single",
  "stem": "Which king broke away from the Church of Rome?",
  "options": [
    {"label": "Henry VIII", "correct": true},
    {"label": "Edward I", "correct": false},
    {"label": "Charles II", "correct": false},
    {"label": "James I", "correct": false}
  ],
  "explanation": "Henry VIII set up the Church of England when the Pope would not let him end his first marriage.",
  "handbook_ref": "Chapter 3 · The Tudors",
  "difficulty": 2
}
```

Automatic checks (validate.py)

- Schema valid; ID format `Q-<chapter>-<section>-<nnn>`; IDs unique across all releases.
- Correct-answer count matches type (1, 2, or true/false); 4 options for single and multi.
- No duplicate or near-duplicate stems (normalised text similarity above 0.9 flagged).
- Stem ≤ 160 characters, option ≤ 70 , explanation 40–300 characters; no "all of the above".
- Readability check on explanations (flag anything above about B2 level).

Review and build

1. `validate.py` exports passing drafts to a Google Sheet: one row per question with columns Status (approve, edit, reject), Edited text, Reviewer note.
2. Amila reviews against the official handbook. Nothing is copied from the handbook or other prep sites: all wording is original.
3. `build_db.py` imports approved rows, writes `content.db`, sets `meta.content_version`, runs integrity checks and writes a changelog.
4. `tts.py` generates audio for new or changed questions only (section 11).
5. The new `content.db` is committed to `app/src/main/assets/`; CI builds and releases (section 15).

11 Audio

Item	Specification
What is voiced	Question stem plus options ("A: ..., B: ..."), explanations, and each section note.
Generation	Pre-generated at build time with a high-quality TTS service (for example ElevenLabs), British English voice, then normalised with FFmpeg.
Format	Opus in Ogg, mono, 24 kbps, loudness –16 LUFS. Roughly 45–60 KB per 15-second clip.
Delivery	Play Asset Delivery pack <code>audio_pack</code> , fast-follow: downloads in the background right after install.
Fallback	If the pack is not ready or a clip is missing, use Android TextToSpeech with an en-GB voice.

Item	Specification
Incremental	<code>tts.py</code> hashes the text; only new or changed text is re-voiced.
Size budget	Audio pack under 80 MB for 1,000 questions plus notes.

12 Firebase on the free Spark plan

Why Spark is enough. Cloud Storage is the only Firebase product here that would require the paid Blaze plan, and this design does not use it. Firestore's free quota is 50,000 reads, 20,000 writes and 20,000 deletes per day with 1 GiB stored, which covers thousands of daily users when each session makes one write. On Spark, exceeding a quota pauses requests until midnight Pacific time; it never creates a bill, and the offline-first app keeps working.

12.1 Projects

Project	Used by	Firestore location	Android package
litu-dev	Debug builds, internal testing	europe-west2 (London)	<code><package>.dev</code>
litu-prod	Closed testing, production	europe-west2 (London)	<code><package></code>

Create the Firestore database in Standard edition, default database, location `europe-west2`. The location cannot be changed later.

12.2 Authentication

- On first launch, sign in anonymously. This gives every install a `uid` for reports and backup without asking for anything.
- "Back up my progress" in Settings links a Google account to the anonymous user (`linkWithCredential`). On a new phone, signing in with Google restores the backup.
- Register debug and release SHA-1 and SHA-256 fingerprints, including Play App Signing's key, for Google sign-in.

12.3 Backup sync design

Progress is backed up as **one document per user**, written once at the end of each study session. Attempts stay on the phone; only the state needed to restore is synced.

```

users/{uid}                                // one document, well under 1 MiB
{
  schemaVersion: 1,
  updatedAt: Timestamp,
  settings:    { testDate, dailyGoal, reminderTime },
  reviewState: { "Q-CH3-TUD-014": [ease, intervalDays, dueAt, reps, lapses, flagged,
updatedAt], ... },
  mocks:      [ { id, finishedAt, score, passed, breakdown }, ... ],    // last 30
  dailyStats: { "2026-10-01": [answered, correct, minutes], ... },      // last 120 days
  streak:     { current, best, lastDay }
}

reports/{autoId}                           // question reports, create-only
{ uid, questionId, reason, comment, contentVersion, appVersion, createdAt }

```

- **When:** a WorkManager job runs after each session ends, only for users with a linked Google account and network.
- **Merge:** on restore or when the remote doc is newer, merge per question by `updatedAt` (latest wins); mocks and daily stats are unioned by ID or day.
- **Cost:** one read on app start (signed-in users only) and one write per session.

12.4 Security rules

```

rules_version = '2';
service cloud.firestore {
  match /databases/{db}/documents {
    match /users/{uid} {
      allow read, write: if request.auth != null && request.auth.uid == uid
                        && request.resource.size() < 900 * 1024;
    }
    match /reports/{id} {
      allow create: if request.auth != null
        && request.resource.data.uid == request.auth.uid
        && request.resource.data.reason in ['wrong', 'unclear', 'typo', 'other']
        && request.resource.data.comment.size() <= 500;
      allow read, update, delete: if false;
    }
  }
}

```

Enable **App Check** with the Play Integrity provider and enforce it on Firestore once the production build is live.

12.5 Remote Config keys

Key	Type	Purpose and default
<code>min_version_code</code>	number	Force-update screen below this version. Default 1.
<code>hidden_question_ids</code>	JSON array	Hide faulty questions until the next content release. Default <code>[]</code> .
<code>banner_message</code>	string	Optional message on Home (for example "We've corrected 3 questions, please update"). Default empty.

Key	Type	Purpose and default
<code>mock_chapter_weights</code>	JSON map	Override mock allocation per chapter. Default empty (proportional).
<code>free_mock_count</code>	number	Mocks available without a subscription. Default 1.
<code>paywall_variant</code>	string	Switch between paywall layouts for testing. Default <code>"a"</code> .

12.6 Analytics events

Event	Parameters
<code>onboarding_complete</code>	<code>has_test_date</code> , <code>daily_goal</code>
<code>sample_quiz_complete</code>	<code>score</code>
<code>paywall_view</code>	<code>source</code> (onboarding, locked_item, settings), <code>variant</code>
<code>trial_start</code> , <code>purchase</code>	<code>product_id</code> (from RevenueCat callbacks)
<code>practice_session_complete</code>	<code>mode</code> , <code>answered</code> , <code>correct</code> , <code>duration_s</code>
<code>mock_complete</code>	<code>score</code> , <code>passed</code> , <code>duration_s</code>
<code>review_session_complete</code>	<code>answered</code> , <code>correct</code>
<code>question_reported</code>	<code>question_id</code> , <code>reason</code>
<code>backup_linked</code>	—

No names, emails or free text in analytics. Analytics data retention set to the minimum that still supports funnel analysis.

13 Subscriptions and entitlements

- Configure products in Play Console (section 4), then mirror them in RevenueCat with one entitlement, `pro`, and one offering, `default`, containing monthly and annual packages.
- The app identifies the RevenueCat customer with the Firebase `uid`, so entitlements follow the user when they link Google.
- `EntitlementRepository` exposes `isPro: StateFlow<Boolean>`. It caches the last known state so a user with no signal keeps access; RevenueCat's own cache handles grace periods.
- Paywall shows live prices from the offering (never hard-coded), trial length, renewal period and "Cancel anytime in Google Play".
- Restore purchases button on the paywall and in Settings.
- Test with Play licence testers on the internal track before closed testing.

14 Non-functional requirements

Area	Requirement
Minimum device	Android 8.0 (API 26), 2 GB RAM; reference low-end test device Android 10 with 3 GB RAM.
Install size	Base app under 20 MB download; audio pack under 80 MB; total on device about 70–100 MB.
Cold start	Under 1.5 s on a mid-range phone, under 2.5 s on the low-end reference device.
Interaction	Next question renders in under 100 ms; no dropped frames on answer feedback animation.
Memory	Under 150 MB in normal use; load one question at a time, never the whole bank.
Offline	Every study feature works in airplane mode. Network is needed only for install, purchase, sign-in and backup.
Battery	No background work except one daily reminder and post-session sync.
Stability	Crash-free users above 99.5% (Crashlytics) before widening a rollout.
Accessibility	WCAG 2.2 AA; works at 200% font size; full TalkBack support (section 23).
Security	No API keys or secrets in the app beyond Firebase config; App Check on; HTTPS only; R8 obfuscation on release.
Privacy	Data minimisation: no name, email only if the user links Google; account and data deletion in-app.

15 Build, CI/CD and release

Build variants

Variant	Firebase project	Application ID	Play track
devDebug	litu-dev	<package>.dev	— (local)
devRelease	litu-dev	<package>.dev	Internal testing (separate app listing)
prodRelease	litu-prod	<package>	Closed testing → Production

GitHub Actions

- **ci.yml** on every pull request: lint, detekt, unit tests, Room migration tests, content validation (`validate.py` on the committed content.db), debug build.
- **release.yml** on a version tag: build signed `prodRelease` AAB with the audio pack, upload to Play internal testing with Gradle Play Publisher.
- Upload key stored in GitHub Actions secrets; Play App Signing holds the app signing key. Keep an offline backup of the upload key.

Release checklist

1. Bump `versionCode` and `versionName` ; update content changelog if content changed.

2. Internal track smoke test on the low-end and mid-range phones (onboarding, purchase with licence tester, mock exam, offline mode).
3. Promote the same bundle to closed testing; check the Play pre-launch report.
4. Production staged rollout 10% → 50% → 100%, watching Crashlytics for 48 hours at each step.
5. If a published question is wrong: add its ID to `hidden_question_ids` immediately, fix it in the next release.

16 Testing strategy

Level	What is tested
Unit	Scheduler, readiness, mock builder, streak logic, sync merge, entitlement caching. Target 90% coverage of <code>core/domain</code> .
Database	Room DAO tests on both databases; every <code>user.db</code> migration tested from every previous version; content upgrade keeps progress.
Content	<code>validate.py</code> in CI: schema, answer counts, unique IDs, no deleted IDs, audio keys resolve.
UI	Compose tests for onboarding → payroll, practice question and feedback, full mock exam, review queue.
Screenshot	Roborazzi screenshots of key screens in light, dark and 200% font size to catch layout breaks.
Firestore rules	Rules unit tests with the Firebase Emulator Suite.
Performance	Macrobenchmark for cold start and question scrolling; Baseline Profile generated in CI.
Manual	Release checklist (section 15) plus TalkBack pass on each major release.

17 Delivery plan

About six weeks of evening and weekend work, with a gate at the end of each two-week phase. The next phase starts only when the gate is met.

Phase	Amila (content, product)	Yasantha (app, backend)	Gate
Weeks 1–2 Foundations	Sign partnership agreement; syllabus map; generate ~1,000 drafts; set up review sheet	Project setup and modules; both Room databases; content import; Firebase dev project; practice screen	Drafts generated; app shows real questions from content.db
Weeks 3–4 Core features	Review and approve 600+ questions; write section notes; store listing copy	Mocks, review queue, readiness, notes, timer, reminders, anonymous auth, backup sync	Content v1 frozen; all core screens working offline
Weeks 5–6 Polish and launch	Audio generation; recruit 12+ UK testers; fix reported questions; launch marketing	Paywall and RevenueCat; audio pack; App Check; CI release; closed testing; production rollout	Live on Google Play

Part C – Design

Everything a designer needs to produce the Figma file: brief, brand, tokens, flows, screen designs, screen specifications and handoff checklist.

Design canvas. All 26 screens in light and dark, plus the logo options, are on the design canvas "Life in the UK App – Screens" (claude.ai/artifact/XyMNXuSQpVPR62Pbo2m5J7). It is private until Amila shares it. Where this document and the canvas differ, this document wins.

18 Design brief

Design a calm, confident study coach for nervous adults, many of whom speak English as a second language and use budget Android phones.

Persona	Profile
Priya, 34 (primary)	Nurse on a skilled worker visa, applying for settlement. Studies in 10–15 minute bursts on the bus. Reads English well but slowly. A failed attempt costs her £50 and weeks of delay.
Tomasz, 41	Warehouse supervisor applying for citizenship. Prefers listening to reading; studies in the evening with audio. Uses dark mode at night.

Design principles

- Calm over gamified.** Progress and encouragement, not flashing rewards.
- One job per screen.** A question screen shows one question and nothing competing with it.
- Plain English.** Short sentences, no idioms; explanations readable at about B1 level.
- Thumb-first.** Answers are large full-width cards in the lower two-thirds of the screen.
- Always know where you stand.** Readiness and today's progress visible from Home.
- Works offline.** No spinners in the study flow.

Brand direction

- **Feel:** trustworthy, modern, warm and a little colourful – a friendly tutor, not a government form.
- **Logo:** a clock tower rising from an open book (section 19.4). It says "UK" and "study" without any official symbol.
- **Colour:** deep teal as the base, with amber, coral and plum accents taken from the logo. Each handbook chapter has its own colour (19.3).
- **Avoid:** Union Jack (including recoloured versions), crowns, royal arms, the Parliament portcullis, red-white-blue flag palettes, GOV.UK styling, and trademarked UK icons such as the red phone box and the Underground roundel.
- **Illustration:** simple line drawings of everyday UK life (bus stop, library, quiet study corner) in teal with amber details, on onboarding, empty states and dialogs only.
- **Tone of voice:** encouraging and direct: "You're 4 questions from today's goal", not "Great job, superstar!"

19 Design tokens

Built on Material 3. Every text and background pair must meet WCAG AA contrast (4.5:1 for body text). Hex values are proposals for the designer to refine and verify; the contrast ratios quoted below were checked for v1.1.

19.1 Colour

Token	Light	Dark	Use
<code>primary</code>	 #0F6E6E	 #6FD3CF	Buttons, selection, progress
<code>onPrimary</code>	 #FFFFFF	 #00302F	Text on primary
<code>primaryContainer</code>	 #E3F0EF	 #124F4E	Selected cards, highlights, chapter 2 tile
<code>secondary</code>	 #3D5A80	 #A9C3E8	Secondary actions; chapter 5
<code>accent</code>	 #E9A23B	 #F5C57A	Streaks, flags; never text on white
<code>success</code>	 #1E7B4F	 #7FD8A8	Correct, passed
<code>successContainer</code>	 #E4F2EA	 #173A2A	Correct answer card, Pass badge NEW
<code>error</code>	 #B3261E	 #F2B8B5	Wrong, failed
<code>errorContainer</code>	 #F9E3E1	 #3F1F1D	Wrong answer card, Fail badge NEW
<code>warning</code>	 #8A5A00	 #F5C57A	Timer under 5 minutes; tip text
<code>background</code>	 #FBF8F2	 #101414	App background (light was #F8FAF9)
<code>surface</code>	 #FFFFFF	 #1A2020	Cards, sheets
<code>surfaceVariant</code>	 #ECEFECE	 #2A3331	Unselected answers, inputs (light was #E6EEEC)
<code>textPrimary</code>	 #1A1F1E	 #E7EDEB	Headings, body
<code>textSecondary</code>	 #4F5A58	 #B5C1BE	Captions, hints
<code>outline</code>	 #D3D6CF	 #3E4846	Borders, dividers (light was #C3CDCA)

Never use colour alone for right and wrong: always pair it with a tick or cross icon and a text label.

19.2 Brand colours **NEW**

These come from the logo and give the app its warmth. Coral and amber are fills and icons only: never use them as text on `background` or `surface`.

Token	Light	Dark	Use
<code>brand</code>	 #0F6E6E	 #0F5E5E	Brand cards: splash, welcome hero, readiness card, new-mock card
<code>onBrand</code>	 #FFF4E0	 #FFF4E0	Headings on brand cards (cream) – 5.5:1

Token	Light	Dark	Use
onBrandMuted	 #E3F0EF	 #E3F0EF	Secondary text on brand cards – 5.2:1
brandAmber	 #FFC24B	 #FFC24B	Readiness ring, "Test in" chip, Start button on brand cards
onBrandAmber	 #16323A	 #16323A	Text on brandAmber – 8.4:1
brandCoral	 #F2765A	 #F2765A	Fills and icons only – never text
coralContainer	 #FDE7E1	 #3D221B	Review tile, review icon, avatar
onCoralContainer	 #B5452C	 #F4A08A	Text and icons on coralContainer – 4.6:1 / 7.1:1
amberContainer	 #FFF1D6	 #3A2E14	Mixed tile, tips, streak chip (text in warning)
plum	 #6A4C9C	 #B49BE0	Chapter 1; Mock test shortcut icon
plumContainer	 #EEE8F6	 #2B2340	Chapter 1 tile, Mock test shortcut
onPlumContainer	 #6A4C9C	 #C9B5EE	Icons on plumContainer – 5.6:1 / 8.0:1
blueContainer	 #E4EAF3	 #1E2A3B	Chapter 5 tile
streak	 #E07B00	 #F5A04A	Streak flame icon only

19.3 Chapter colours NEW

Each handbook chapter has one colour so learners recognise it at a glance. The colour marks the chapter's number tile (container fill with a 4 dp base line in the chapter colour), its mastery and results bars, and the dot on weakest-topic chips. It is always shown next to the chapter number or name, never alone and never as text colour.

#	Chapter	Colour (light / dark)	Tile container
1	Values and principles	 #6A4C9C /  #B49BE0	plumContainer
2	What is the UK?	 #0F6E6E /  #6FD3CF	primaryContainer
3	History	 #F2765A /  #F2765A	coralContainer
4	Modern society	 #E9A23B /  #E9A23B	amberContainer
5	Government and law	 #3D5A80 /  #A9C3E8	blueContainer

19.4 Logo and app icon NEW

The logo is a clock tower with a coral spire and an amber clock face, rising from an open book with one amber and one coral page, on a deep teal tile. It is a generic clock tower silhouette, not a reproduction of any official building mark.



Item	Rule
Colours	Tile #0F6E6E ; tower and clock surround cream #FFF4E0 ; clock face and left page amber #FFC24B ; spire and right page coral #F2765A ; clock hands ink #16323A . The logo never changes colour between light and dark themes.
Adaptive icon	108 × 108 dp grid. Background layer: flat #0F6E6E . Foreground layer: tower and book, scaled so the whole drawing sits inside the 66 dp safe circle (the canvas concept is drawn larger and must be scaled down).
Themed icon	One-colour silhouette of tower and book for Android 13+ themed icons, with the window lines and clock hands cut out.
In the app	Splash (200 dp, no tile, on brand), Welcome hero, Paywall (56 dp), Home header (44 dp). A faint 16% opacity tower decorates the readiness and new-mock brand cards.
Don't	Add a flag, crown or red-white-blue; recolour the parts; put text inside the tile; rotate or stretch it.

19.5 Typography

Style	Font	Size / line height (sp)	Weight	Use
Display	Plus Jakarta Sans	32 / 40	700	Readiness and results scores
Headline	Plus Jakarta Sans	24 / 32	700	Screen titles
Title	Plus Jakarta Sans	18 / 26	600	Card titles, section headers
Question	Atkinson Hyperlegible	20 / 30	700	Question stem
Body	Atkinson Hyperlegible	16 / 24	400	Answers, explanations, notes
Label	Atkinson Hyperlegible	14 / 20	700	Buttons, chips, tabs
Caption	Atkinson Hyperlegible	12 / 16	400	Metadata, hints

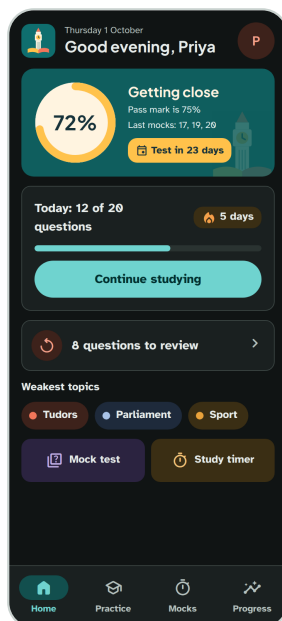
Atkinson Hyperlegible was designed for letter clarity, which helps readers whose first language is not English. Both fonts are free (Google Fonts) and bundled in the app, not downloaded.

19.6 Spacing, shape, elevation, motion

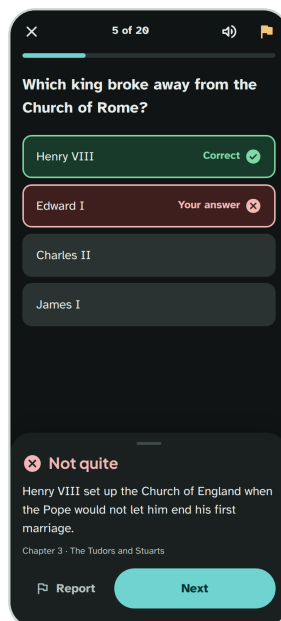
Token	Value
Spacing scale	4, 8, 12, 16, 24, 32, 48 dp
Screen margin	16 dp phone, 24 dp tablet
Minimum touch target	48 × 48 dp; 8 dp between answer cards
Corner radius	8 dp chips · 12 dp cards and answers · 16 dp brand cards and mode tiles NEW · 28 dp pill buttons and sheets
Elevation	Flat cards with 1 dp outline; elevation only for sheets and dialogs
Motion	150–250 ms ease-out; answer feedback 200 ms; respect "Remove animations"
Icons	Material Symbols Rounded, 24 dp, weight 400; filled variant for the selected tab and status icons

19.7 Light and dark themes **NEW**

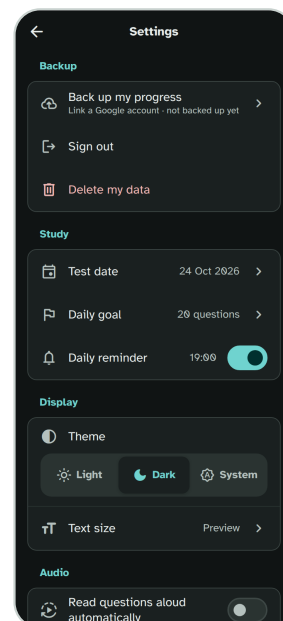
- **Choice:** Light, Dark or System. **System is the default** and follows the phone's dark-theme setting.
- **Where:** a three-way segmented control in Settings > Display (S20). The change applies instantly, without restarting the app or losing the current screen.
- **Storage:** the existing DataStore key `theme` (section 8.4) with values `system` (default), `light`, `dark`.
- **Coverage:** every screen, sheet and dialog is designed in both themes. Dark uses the Dark column of 19.1–19.3; `background` is #101414, not pure black.
- **Fixed in both themes:** the logo, the splash screen and the `brand` cards keep their colours, so the brand looks the same at night.
- **Not in onboarding:** the theme choice is not an onboarding step; System is right for almost everyone.



S07 Home · dark
Brand card unchanged;
containers darken



S11 Question · dark
Correct and wrong keep
icon and label

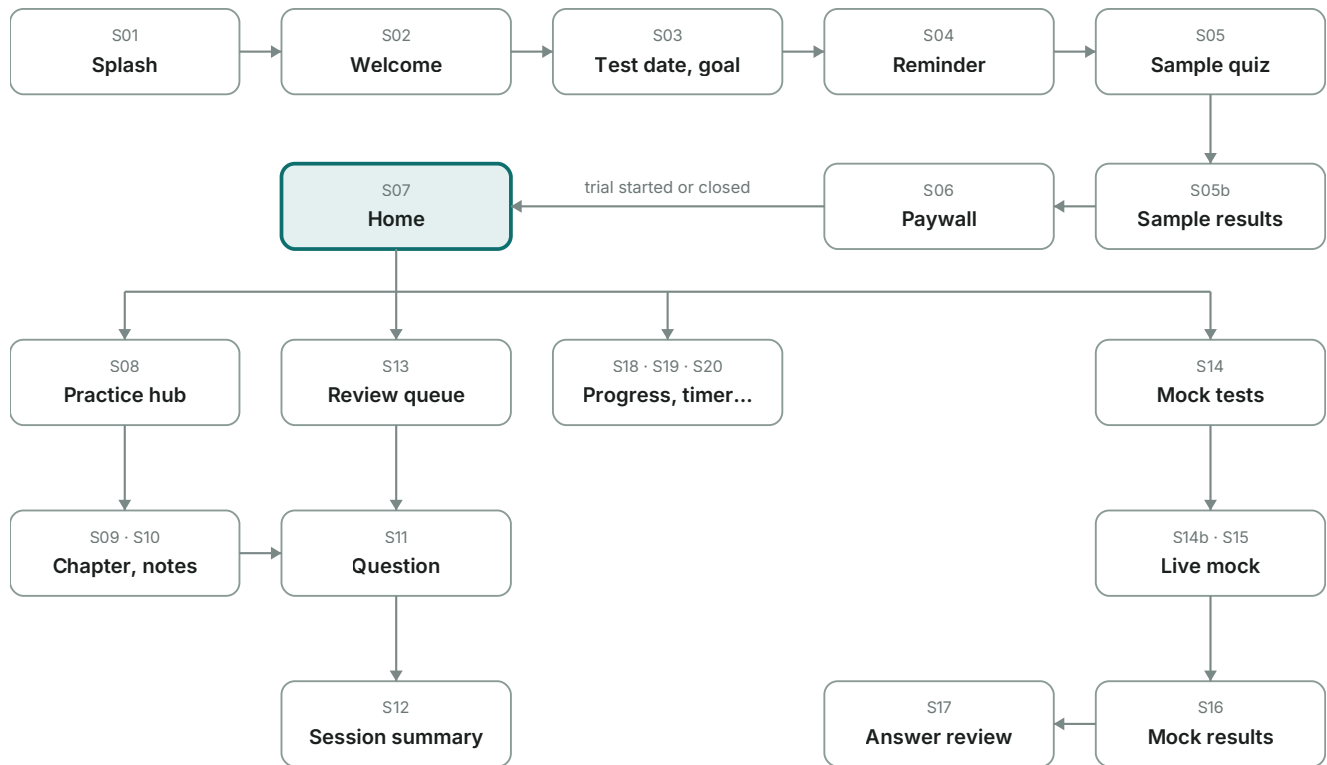


S20 Settings · dark
Theme picker in Display

20 Navigation and user flows

Four bottom tabs – Home, Practice, Mock tests, Progress. Settings opens from the avatar on Home. First launch runs straight through onboarding to a free sample, its results and the paywall.

Onboarding ends at the paywall; Home branches into practice, review and mocks

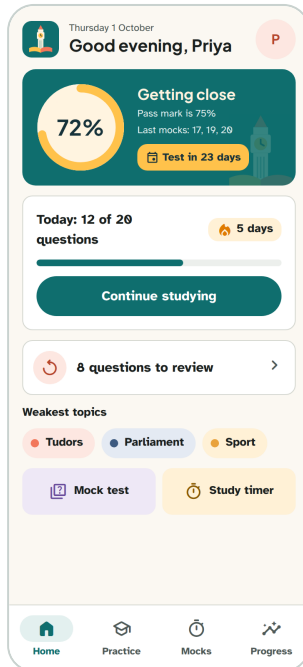


Free users: locked items open the paywall (S06). Summaries and results return to Home. S11b, S13b, S15b and S16b are sheets or states of the screen next to them.

Figure 3 – Screen flow (screen IDs match section 22)

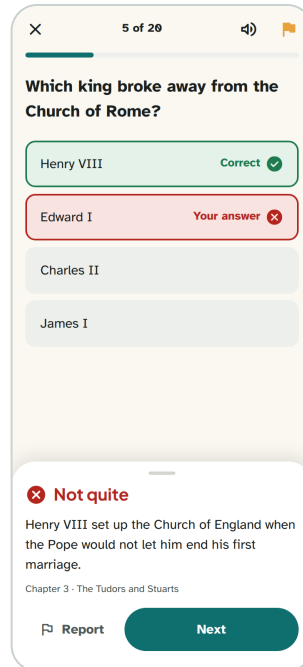
21 Key screen designs

High-fidelity designs of the six most important screens in the light theme, taken from the design canvas. They replace the v1.0 wireframes. The designer refines them within the tokens in section 19.



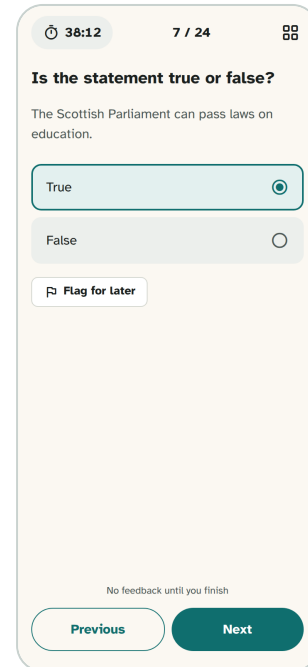
S07 Home

Readiness first, then today's goal



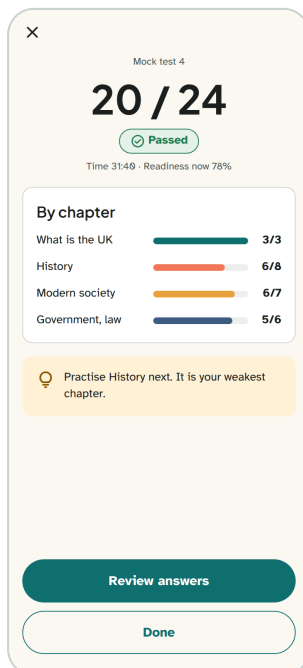
S11 Question – after Check

Answer cards plus feedback sheet



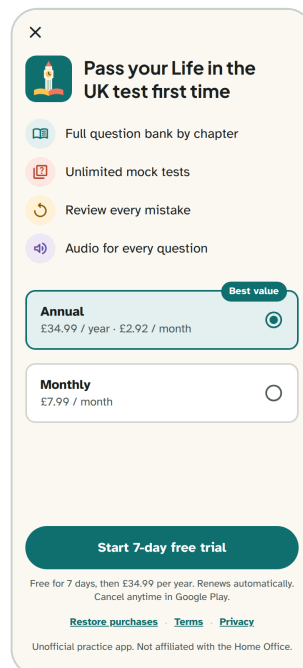
S15 Mock in progress

Exam conditions, timer always visible



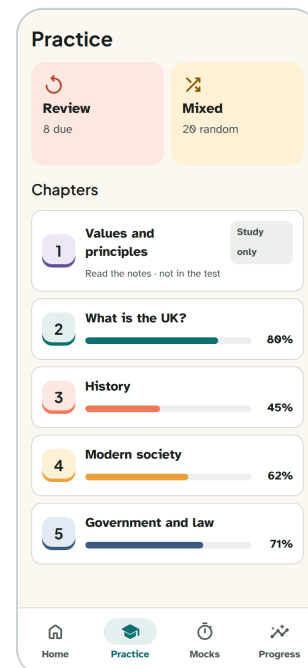
S16 Mock results

Score, pass badge, chapter colours



S06 Paywall

Annual preselected, real prices from Play



S08 Practice hub

Chapter tiles and mastery bars

Sample questions, names and figures in the designs are placeholders for layout only. [APP NAME] is still to be decided.

22 Screen specifications

Twenty screens plus six supporting screens (IDs ending in "b"). "Locked" means a free user sees a lock icon and is sent to the paywall.

ID	Screen	Layout and content	States and rules
S01	Splash	Clock tower logo (about 200 dp, no tile) centred on <code>brand</code> ; app name in <code>onBrand</code> ; tagline "Your Life in the UK study coach"; thin four-colour strip (amber, coral, plum, cream) at the bottom.	Under 1 s. Routes to S02 (first launch), S06 (trial ended) or S07. Same in both themes.
S02	Welcome	3 slides on a <code>brand</code> hero with the clock tower scene, headline, one line, dots, "Continue", "Skip". (1) Pass first time: 24 questions, 45 minutes, 18 to pass. (2) Study in short bursts. (3) Know when you're ready.	Footer: "Unofficial practice app. Not affiliated with the Home Office."
S03	Test date and goal	Date field, "I haven't booked yet" checkbox, three goal cards: 10 (5 min), 20 (10 min), 40 (20 min).	With a date, show "That's 23 days – about 20 questions a day gets you ready" and preselect a goal.
S04	Reminder	Time picker (default 19:00), quick chips (08:00, 12:30, 19:00), "Turn on reminders", "Not now".	System notification permission is requested only after "Turn on reminders".
S05	Sample quiz	S11 layout with header "Free sample · 3 of 10".	Ends with S05b, then S06.
S05b 	Sample results	Score out of 10, one-line message naming the weakest chapter, bars by chapter in chapter colours, tip card, "See plans".	Always followed by S06.
S06	Paywall	Close (X); logo (56 dp) beside the headline; 4 benefits with icons in four tinted circles; plan cards (annual preselected with "Best value"); "Start 7-day free trial"; small print; Restore, Terms, Privacy.	Loading: skeleton plan cards. Billing unavailable: message and Retry. Already subscribed: go to S07. Prices always from Play.
S07	Home	Logo (44 dp), date and greeting, avatar; readiness card on <code>brand</code> with amber ring and "Test in 23 days" chip; today card with streak chip, progress and "Continue studying"; review card; weakest-topic chips with chapter-colour dots; Mock test and Study timer shortcuts on plum and amber containers; optional Remote Config banner.	Before first mock, readiness card says "Take a mock test to get your score".
S08	Practice hub	Review tile (coral) and Mixed tile (amber); list of 5 chapters, each with a numbered tile in its chapter colour, name, mastery bar in the chapter colour and %.	Chapter 1 labelled "Study only". Locked for free users except sample content.
S09	Chapter detail	Title, mastery %, "Practise this chapter", "Read notes"; sections with own mastery bar and count.	Tap a section to practise only that section.

ID	Screen	Layout and content	States and rules
S10	Study notes	Original summary in short paragraphs; key-fact boxes (dates, names, numbers); audio button; "Test yourself on this section".	Footer link to buy the official handbook.
S11	Question (practice)	Close, "5 of 20", progress bar, audio, flag; stem; 2–4 answer cards; pinned "Check" button. After Check: cards marked with icon and label ("Correct", "Your answer"), bottom sheet with verdict, explanation, handbook reference, "Next", "Report".	"Choose 2" hint for multi questions. Long stems scroll; button stays pinned. No swipe-to-answer.
S11b NEW	Report a question	Bottom sheet over S11: question ID, four reasons (answer wrong, unclear, spelling, something else), optional note, "Cancel", "Send report".	Works offline; queued and sent with the next backup sync.
S12	Session summary	"16 of 20", short message, missed questions list, goal progress, streak, "Practise again", "Done".	Triggers backup sync for linked users.
S13	Review queue	Due count, estimated minutes, "Start review" on an amber container; "Coming up" list; then S11 per question.	Empty state is S13b.
S13b NEW	Review queue, empty	Line illustration, "Nothing to review", short line, "Mixed practice" button.	Shown when no items are due.
S14	Mock tests	"Mock test 5" card on brand (24 questions · 45 min · 18 to pass) with an amber "Start mock test" button; history list with date, score, Pass/Fail badge.	Mock 1 free.
S14b NEW	Start confirmation	Dialog: illustration, "Ready to start?", three rules (quiet place, 45 minutes and no pause, no feedback until the end), "Not yet", "Start".	Shown every time a mock starts.
S15	Live mock	Timer, "7 / 24", navigator icon; question without feedback; Previous, Next, Flag.	Timer turns warning colour under 5 min; auto-submits at 0.
S15b NEW	Navigator sheet	24 squares in 6 columns (answered, not answered, flagged, current), legend, warning about unanswered questions, "Back to test", "Finish test".	"Finish test" warns when any question is unanswered.
S16	Mock results	Score, Pass/Fail badge (icon + text), time, chapter bars in chapter colours, updated readiness, next-step tip, "Review answers", "Done".	Fail state is S16b.
S16b NEW	Mock results, not passed	Score, "Not passed" badge, "You needed 18 to pass", supportive card ("You were 2 questions away"), "Practise your 2 weakest chapters", "Review answers".	Never blames; always offers the next step.
S17	Answer review	Filter chips All / Wrong / Flagged; expandable list with chosen and correct answers and explanation.	Wrong answers added to the review queue.
S18	Progress	Readiness over time (line chart with 75% pass line), streak, study minutes this week, total answered, mastery by chapter in chapter colours, mock history.	Charts need text alternatives for TalkBack.

ID	Screen	Layout and content	States and rules
S19	Study timer	Large circular timer (default 15 min), start/pause, reset, skip, length chips 10/15/25.	At the end: "Quick 5-question quiz?" opens S11.
S20	Settings	Backup (link Google, sign out, delete data); Study (date, goal, reminder); Display (theme picker: Light / Dark / System , text size preview); Audio (autoplay, speed); Subscription (manage, restore); About (version, content version, disclaimer, handbook link, privacy, terms, support).	Theme defaults to System and applies instantly (19.7). Delete data removes user.db, Firestore doc and Auth user after confirmation.

23 Components, accessibility and handoff

23.1 Components (design every state, in light and dark)

Component	States
Answer card	Default, pressed, selected, correct, wrong, correct-not-chosen, disabled; single and "choose 2" variants
Buttons (primary, secondary, text)	Default, pressed, disabled, loading; amber variant on brand cards
Feedback sheet	Correct, wrong, long explanation (scrolls), with Report action
Readiness ring	No mock yet, below 60%, 60–79%, 80%+ (on the brand card)
Mastery bar	0%, partial, 100% (mastered tick); one colour per chapter
Chapter tile NEW	Chapters 1–5 in their colours; "Study only"; locked
Brand card NEW	Readiness card and new-mock card, with faint logo decoration
Theme picker NEW	Light, Dark or System selected; pressed; focus
Countdown timer	Normal, under 5 minutes, expired
Navigator square	Unanswered, answered, flagged, current
Plan card	Default, selected, "Best value" badge, loading skeleton
Chips, badges	Chip default/selected; chapter-dot chip; Pass and Fail badges; streak and freeze chips
Empty and error states	Review empty (S13b), no mocks, billing unavailable, audio not downloaded
Bottom navigation	4 tabs, each selected; lock badge for free users

23.2 Accessibility (WCAG 2.2 AA)

- Contrast at least 4.5:1 for text and 3:1 for icons and answer-card outlines, in both themes.
- Coral and amber are never used as text on **background** or **surface**; chapter colours always sit next to the chapter number or name.

- Touch targets at least 48 × 48 dp, 8 dp between answer cards.
- Layouts hold at 200% system font size: nothing truncated, buttons wrap or grow.
- Right and wrong shown by icon and text, never colour alone.
- Every icon button has a TalkBack label; the mock timer announces every 5 minutes, not every second.
- Audio read-aloud for every question and note; light and dark theme for every screen, following the system setting by default.

23.3 Designer handoff checklist

- Figma file with pages: Foundations (tokens), Components, Screens, Prototype; styles named exactly as the tokens in section 19, including the brand and chapter tokens.
- All 26 screens (S01–S20 plus S05b, S11b, S13b, S14b, S15b, S16b) in light and dark at 360 × 800 dp; S07, S11, S15 also at 200% font and on a 600 dp-wide tablet.
- All component states from 23.1, in both themes.
- Clickable prototype: onboarding → sample → sample results → paywall → practice → mock → results.
- Final logo artwork as SVG; adaptive app icon (foreground and background layers, 432 × 432 px) and monochrome themed icon, built from the clock tower logo (19.4).
- Play Store icon 512 × 512 px, feature graphic 1024 × 500 px and 6–8 phone screenshots with captions, including at least one dark screenshot.
- Illustrations exported as SVG; motion notes for answer feedback, the bottom sheet and the theme switch.

Part D – Governance

Legal, privacy, risks and the decisions still open.

24 Legal, privacy and compliance

The two biggest legal risks are copying content and looking like an official government app; both are avoided by design. This is not legal advice; confirm these points with a lawyer before launch.

Area	Rule for this app
Handbook content	The handbook is Crown copyright and sold through the official shop. Do not reproduce its text. Write original questions and summaries testing the same facts, and link to the official handbook.
Other prep sites and apps	Never copy questions from competitor websites or apps. All questions are original.
Official status	"Unofficial practice app, not affiliated with the Home Office" in the store listing, on S02 and S06, and in About. No crown, GOV.UK styling or government logos; the app name must not imply official status.
UK GDPR	Public privacy policy; data minimisation; progress backup stored in London; in-app and web account deletion as Google Play requires.
ICO	Check whether the business must pay the ICO data protection fee.
Google Play	Complete the Data safety form; show price, renewal period and cancellation on the paywall; follow Play's subscription policy.
Children	The test is for adults; set the target audience to 18+ in Play Console.

Privacy policy summary for users: progress stays on your phone. If you choose to back up, it is stored in Google's London data centre. Crash reports and anonymous usage statistics are processed by Google. Payments are handled by Google Play. We never sell data.

25 Risks and open decisions

Risk	Impact	Mitigation
Wrong or misleading questions	High	Human review of every question; in-app reporting; Remote Config hide list.
Crowded market, free alternatives	High	Win on experience: offline, readiness score, audio, plain English; strong store listing.
Test changes after the White Paper	Medium	Versioned content.db; new syllabus ships as an update.
Low Android conversion	Medium	Hard paywall, annual plan, UK-focused marketing.
Developer time drops	Medium	Small MVP, weekly 15-minute check-in, clear phase gates.
Partnership disagreement	High	Written agreement signed before building.

Open decisions

#	Decision	Owner	Needed by
1	App name, package name and logo direction (must not look official)	Amila	Start of week 1
2	Partnership agreement: publisher account, revenue split, ownership of code and content, exit terms	Both	Before any code
3	AI model and TTS voice for content and audio	Amila	Week 1
4	Final prices after competitor check	Amila	Week 4
5	Readiness display: ring or horizontal gauge	Designer	Week 2

26 Sources

- GOV.UK – Life in the UK Test: book the test (24 questions, 45 minutes, £50, official handbook): gov.uk/life-in-the-uk-test
- GOV.UK – What happens at the test (75% pass mark, retakes): gov.uk/life-in-the-uk-test/what-happens-test
- uAcademy, Rowan and Connaught Law – 2026 status of the test and White Paper proposals (as of mid to late 2026)
- Firebase – Cloud Firestore locations (europe-west2 London; location cannot be changed): firebase.google.com/docs/firestore/locations
- Firebase – Cloud Firestore billing and free quota: firebase.google.com/docs/firestore/pricing
- Firebase – Cloud Storage billing requirements after September 2024 (Blaze required): firebase.google.com/docs/storage/faqs-storage-changes-announced-sept-2024
- Life in the UK Test Web – example of free competitor content: lifeintheuktestweb.co.uk

Facts about the exam and Firebase pricing were checked on 1 October 2026. Recheck prices, quotas and Play policy before launch.